

Pre-Calibrated Force Program With LoRa Sending To This LoRa Receiving Program that writes Data to SD Card

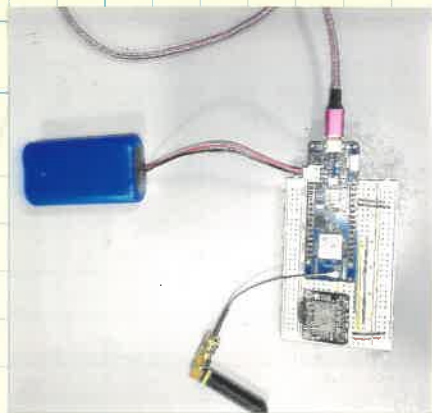
```

#include <SPI.h> // Including libraries
#include <SD.h>
#include <LoRa.h>
const int CS = 11; //Chip Select
float Depth;
File H2FloatFile; //Naming the depth data file

void setup()
{
  Serial.begin (9600);
  while(!Serial); //Must open Serial monitor
  Serial.println("LoRa Receiver");
  if(!LoRa.begin(915E6)) //915 MHz
  {
    Serial.println("Starting LoRa failed");
    while(1); //If LoRa fails, it jams up
  }
  if(!SD.begin(CS)) //If it does not begin sit in the while loop
  {
    Serial.println("SD Failed"); //Print error indicates stuck in while loop
    while(1);
  }
  bool fileExists = SD.exists("H2Float.txt"); //if the file exists this will be true
  if(fileExists > 0) //this will delete the existing file and all its data
  {
    SD.remove("H2Float.txt");
  }
  H2FloatFile = SD.open("H2Float.txt", FILE_WRITE); //Open file will only do this one time at the top
  H2FloatFile.println("New File"); //At very top of the new file
  H2FloatFile.println("_____");
  H2FloatFile.println("Depth"); //Column headers go in this line
  H2FloatFile.close(); //Close file
}

void loop()
{
  float packetSize = LoRa.parsePacket(); //Get the packet
  if(packetSize)
  {
    char rcvd = LoRa.read();
    if(rcvd == 'H') //Packet identifier
    {
      while(LoRa.available())
      {
        Depth = LoRa.parseFloat();
      }
      H2FloatFile = SD.open("H2Float.txt", FILE_WRITE); //Open and Name file
      H2FloatFile.println(Depth); //Put stuff in the file
      H2FloatFile.close(); //Close file
      Serial.println(Depth); //Print SD data To Serial Monitor
    }
  }
}

```



LoRa Receiver with SD card Saving Setup

Data Sent via LoRa to the SD card Saving Receiver → The Data on Serial Monitor and SD card that matches was while the Receiver was Plugged Directly into the Computer For Power. The Section Below where Serial Monitor Data Steps is the Data that Continued to Send while on Battery Power, Only to prove the Setup worked exclusively on Battery Only.

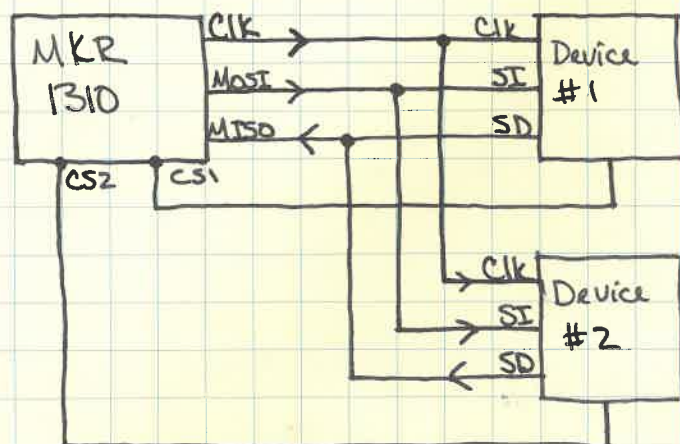
LoRa Receiver Serial Monitor

0.38
0.39
0.40
0.40
0.45
1.51
2.78
4.04
5.03
5.03
5.25
5.24
5.23
5.79
6.90
8.07
9.24
9.91
9.91
10.03
10.19
10.19
10.16
10.10
10.09
10.11
10.11
10.12
10.13
10.14
10.16
10.14
10.11
10.10
10.06

Depth Data Saved to SD Card

0.38
0.39
0.40
0.40
0.45
1.51
2.78
4.04
5.03
5.03
5.25
5.24
5.23
5.79
6.90
8.07
9.24
9.91
9.91
10.03
10.19
10.19
10.16
10.10
10.09
10.11
10.11
10.12
10.13
10.14
10.16
10.14
10.11
10.10
10.06
10.06 Battery Power
10.07
10.09
10.08
10.09
10.05
10.04
10.01

SPI circuit Diagram (Basic)



SDA ⇒ 11
MISO ⇒ 10
MOSI ⇒ 8
SCK ⇒ 9

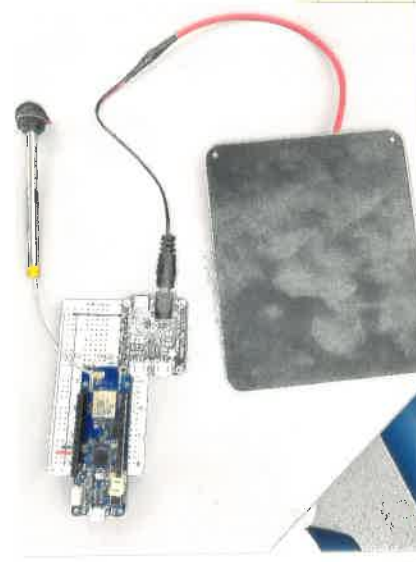
Solar Power Basic Setup / Precalibrated Force Program w/ LoRa Set to be Low Power

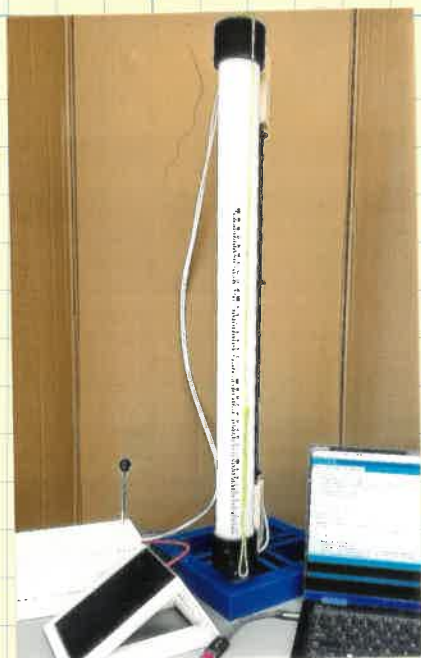
```
#include<Q2HX711.h> //Calls Hx711 Library
#include <SPI.h> //Calls SPI Library
#include <LoRa.h> //Calls LoRa Library
#include <ArduinoLowPower.h> //Calls Low Power Library
Q2HX711 UpperHX711(2,3); //Identifies UpperHX711 pins
Q2HX711 LowerHX711(4,5); //Identifies Lower HX711 pins
long UpperTare = 8598996; //Precalibrated Values (4 lines)
long LowerTare = 8425027;
long UpperWeightedTare = 8805932;
long LowerWeightedTare = 8060228;

void setup()
{
  Serial.begin(9600);
  Serial.println("Starting loop...");
}

void loop() //Starts LoRa in the loop Each Cycle and then turns off
{
  if(!LoRa.begin(915E6)) //915 MHz
  {
    Serial.println("Starting LoRa failed");
    while(1); //If LoRa fails, it jams up
  }
  long UpperReading = UpperHX711.read(); // Each time Upper load cell takes reading
  long LowerReading = LowerHX711.read(); // Each time Lower load cell takes reading
  float MassUpper = 0.75*(UpperReading-UpperTare)/(UpperWeightedTare-UpperTare);
  float MassLower = 0.75*(LowerReading-LowerTare)/(LowerWeightedTare-LowerTare);
  float Force = (MassUpper + MassLower)*9.8/2;
  float Depth = (Force/18.5)*100.0; // Distance of water level rise in cm
  Serial.println(Depth); //Print to verify packet
  LoRa.beginPacket(); //Creates packet to send
  LoRa.print('H'); //Header to Specify H2Float Data
  LoRa.print(Depth); //Data to send in packet
  LoRa.endPacket(); //Closes packet
  delay(10000); //Not needed real world data will be readable on
                //(Patrick's CurRex) current monitor

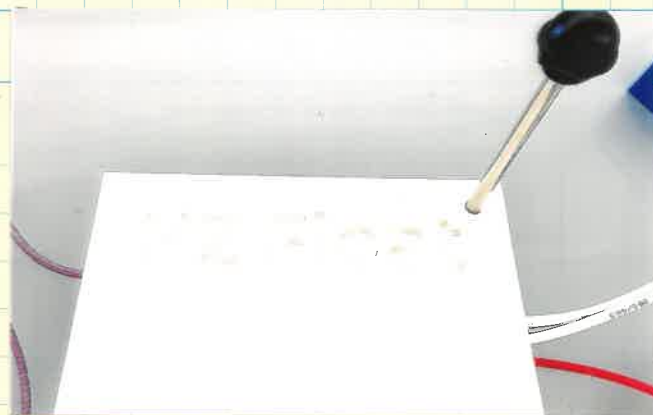
  //Shut down LoRa
  digitalWrite(LORA_RESET, LOW); //Take it low to conserve power
  digitalWrite(LORA_BOOT0, LOW);
  pinMode(LORA_RESET, INPUT);
  pinMode(LORA_DEFAULT_SS_PIN, INPUT); //Slave Select
  pinMode(LORA_IRQ_DUMB, INPUT); //Interrupt
  pinMode(LORA_BOOT0, INPUT);
  //Sleep the MKR1310
  //Can put delay here to see the board without LoRa
  //RTC clock still ticks when the board is in sleep mode
  Serial.println("Sleep the Board...");
  LowPower.deepSleep(10000); //How long it is asleep. Can't re-upload during this sleep
                             // mode. Have to double click the reset button.
}
```



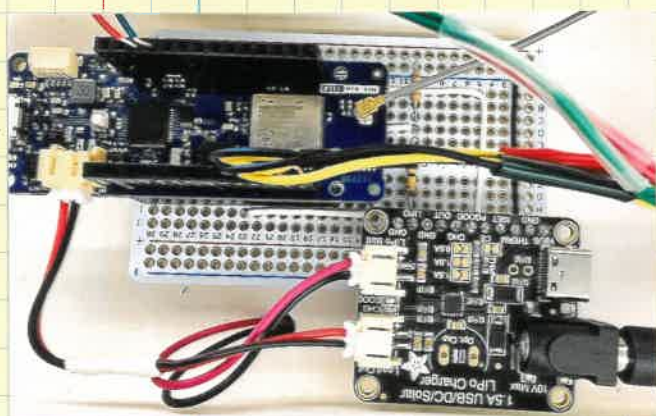
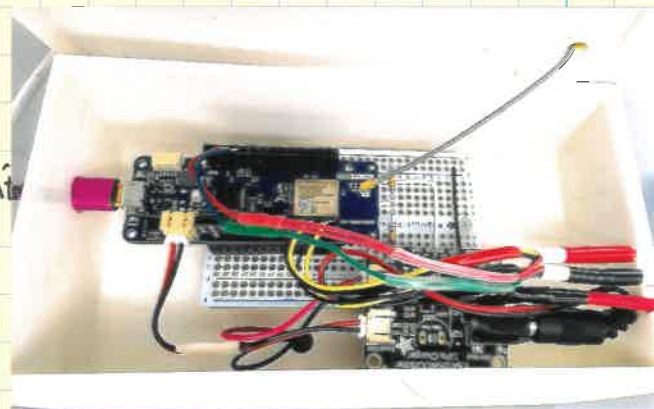


Complete Setup

H2Float →
Components
Containment
allows
Antenna out
Top
Wires Out Side



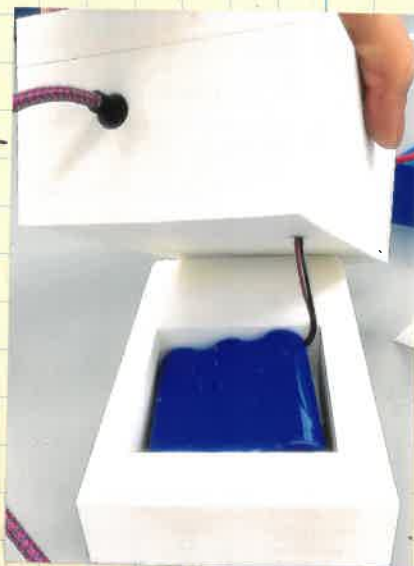
Middle →
Section Electronic
Components



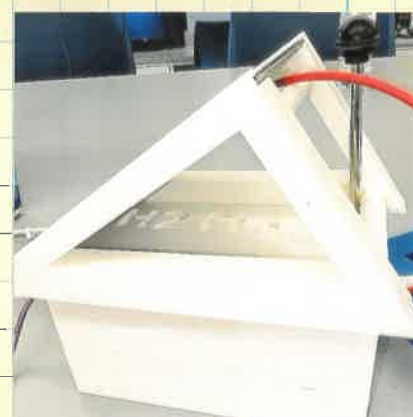
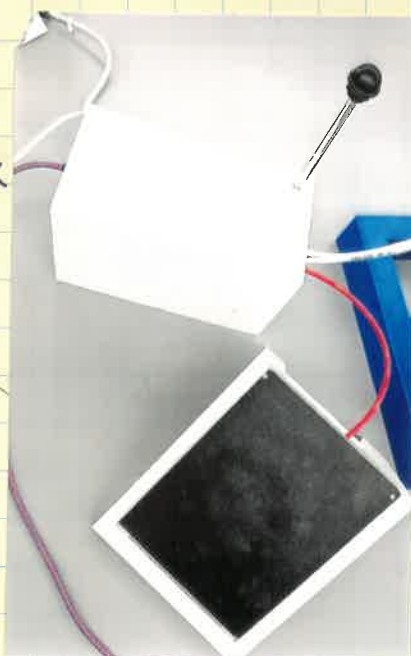
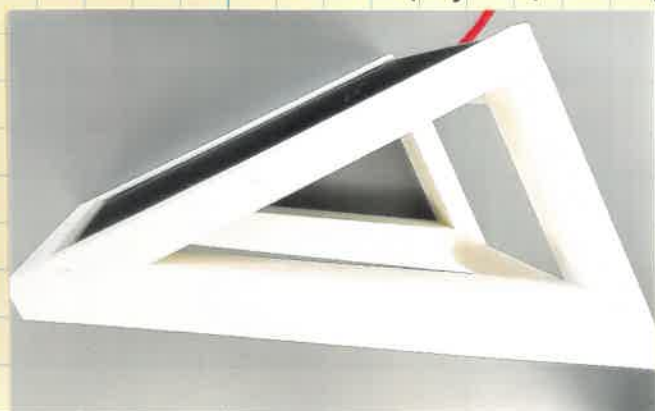
→ MKR1310
Wired for Load Cells
(Attached to Solder Board Via
Stand offs)

→ Solar Power
Battery Charging
For LiPO Battery

Battery
Compartment
on Bottom



Basic.
↓ Cheese
Solar Panel Holder
Fit on top of Component Box



↑ Cheese on top of
Component Box

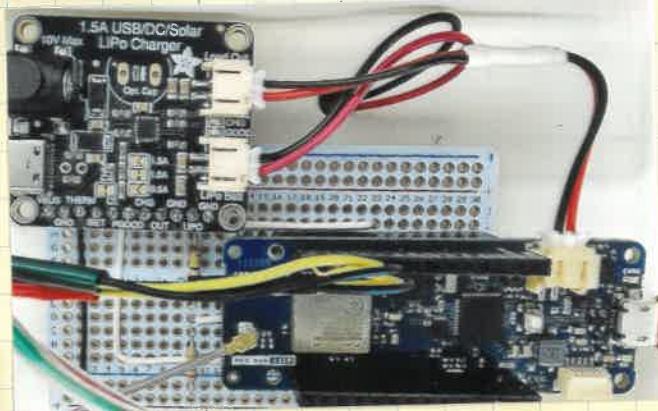
Baffle Construction



- Used Band Saw to Cut In to 24"x9" Pieces
- Hole Saw to make holes for Water Flow 1.5"
- Elbows to hold the Pieces together in 3 Places
- Attached To Floating Pipe Set up On Back 2 sides of Base

J.S.T. Connectors → Japanese Solderless Terminals

- Commonly used on battery Packs (LiPo) for Recharging
- For Device Solar Setup Using AdaFruit BQ24074 Universal USB / DC / Solar Lithium Ion / Polymer Charger 1.5Amp
- 2 JST connector Male Jacks; one for Battery Connection, one for Load out to MKR1310
- MKR1310 1 JST Connector Male Jack Taking Load in for Powering Device.
- LiPO battery Pack 1 Female JST Connector Connects to AdaFruit Solar Charger for Charging the Battery.



RTC For MKR1310

- Must Load RTCZero Library (1.6.0)
#include <RTCZero.h>
- Can be used to keep track of Current time and can be used to program actions at a certain time.

Basic Real Time Clock Program (RTC)

```
#include <RTCZero.h>
#include <SPI.h>
#include <SD.h>
const int CS = 11;
File CoolKidFile;
RTCZero rtc;
const byte seconds = 40;
const byte minutes = 59;
const byte hours = 14;
//const byte days = ;
//const byte months = ;
//const byte year = ;

void setup()
{
  Serial.begin(9600);
  pinMode(CS, OUTPUT);
  rtc.begin();
  rtc.setTime(hours, minutes, seconds);
  //rtc.setDate(day, month, year);
}

void loop()
{
  int h = rtc.getHours();
  int m = rtc.getMinutes();
  int s = rtc.getSeconds();
  if (h < 10)
  {
    Serial.print("0");
  }
  Serial.print(h);
  Serial.print(":");
  if (m < 10)
  {
    Serial.print("0");
  }
  Serial.print(m);
  Serial.print(":");
  if (s < 10)
  {
    Serial.print("0");
  }
  Serial.print(s);
  Serial.println(s);
  delay(5000);
}
```

//Chip select for SD card
//for my SD card stuff
//Create a name (instance) for the rtc stuff
//This will set the current clock time

//This will set the Current Date

//Chip select pin (controls SD SPI) is an output
//initialize RTC 24 hour format
//Sets the starting clock time
//Sets the current date

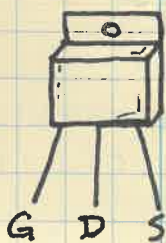
← Utilizes The RTC That is Built into The Arduino
← Does Require Start time to be Set

Metal Oxide
Semiconductor
Field-Effect
Transistor

MOSFET
↓ Blinky Light
Program

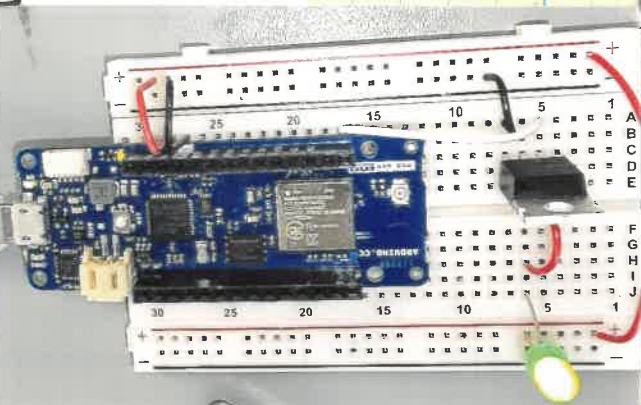
```
void setup()
{
  pinMode(6, OUTPUT); // Controls MOSFET gate
}

void loop()
{
  digitalWrite(6, HIGH);
  delay(500);
  digitalWrite(6, LOW);
  delay(500);
}
```



G - Gate (High/Low)
D - Drain (Ground)
S - Source (Vcc)

Allows one to turn anything on/off using The Transistor Replacing need for a Relay



Outdoor Test #1

6/25 Calibrations Air Temperatures Below 90°F			
Upper Tare	Lower Tare	Upper 750g	Lower 750g
8698817	8525138	8862594	8112531
8700519	8527090	8863603	8113684
8701203	8527885	8860685	8109553
8703803	8530802	8867312	8117195
8705399	8532673	8867856	8115591
8708755	8536350	8871705	8121974
8708436	8536158	8875818	8124977
8703847	8530871	8867082	8116501
3604	4094	4950	5001
0.04	0.05	0.06	0.06

6/25 Calibrations Air Temperatures over 90°F			
Upper Tare	Lower Tare	Upper 750g	Lower 750g
8885941	8723845	9105278	8373563
8900179	8738593	9112420	8381529
8896153	8734102	9107152	8376411
8897250	8735124	9108837	8378235
8900883	8738875	9115704	8385377
8905551	8743595	9116513	8385069
8906509	8744425	9117785	8387242
8898924	8736937	9111956	8381061
6391	6432	4573	4765
0.07	0.07	0.05	0.06

This test run with Respect to Air Temperature Only. Since Both Load cells are Connected One affects the Other. The results Seen here with Air Temperatures above and Below 90°F Show Clearly how the Calibration Factors can Vary with Only the Top Loadcell (out of the 2) Temperature being Varied.

In both Instances the Water Used was from the Same Source with a consistent temperature at output. Over time the temperature would only Vary by about 1 degree(F) for each Calibration Run.

* Thank You Wesley Cook for Helping figure this Out!

Note:

Not All SD Cards are Equal!
There are Some that will Not Work in this Device Set Up. 130 speed is too much and kills the Cam Monitor!



Solar Battery Charging Even in Full Shaded Conditions with Solar Panel Facing Direction of Sunlight The Battery Never dropped below Full Charge. There was more than Enough Solar to Power Device without needing Battery Backup.

This Raises Questions about Viability of Running 2 Load Cells Connected to each Other but in Separate Temperature Conditions.

Outdoor Test #2

Calibration 6/26 Air 84°F : Water 77°F : Start 9:00am				
Upper Tare	Lower Tare	Upper 750g	Lower 750g	Full Shade
8637906	8458016	8899621	8155471	Air 84°F : Water 77°F
8675819	8498235	8900520	8155659	
8675731	8498202	8901531	8157769	
8678624	8501501	8902845	8160289	
8679005	8502051	8903466	8161109	
8680685	8503961	8904971	8163414	
8683354	8506922	8908240	8166930	
8684168	8508107	8909715	8168067	Air 85°F : Water 78°F
8718071	8545079	8963533	8227251	Partial/mostly Shaded
8741253	8569400	8973024	8236460	
8744895	8573263	8967591	8231094	
8735132	8563121	8959745	8222830	
8729136	8556994	8958081	8221664	
8734558	8562815	8971363	8236060	Full Sun
8766526	8646615	9015304	8281734	
8794988	8626522	9058222	8327003	
8875624	8711426	9161609	8435866	Air 87°F : Water 79°F
8791037	8623952	9019350	8282612	Umbrella Shade
8761020	8591964	8992586	8256558	
8746049	8576342	8981442	8243091	
8739264	8569271	8976464	8238551	
8738031	8567999	8980217	8243738	Air 88°F : Water 79°F
8983428	8247471	8985804	8250491	
8754414	8585435	9004848	8268723	
8775942	8608241	9023161	8286247	
8808760	8643034	9058149	8323473	
8822099	8657225	9069275	8336159	
8837396	8673468	9087952	8355893	Air 88°F : Water 80°F
8851159	8688091	9102421	8371912	
Air 88°F : Water 80°F : Finish 10:32 am				
8753244	8564301	8984174	8247108	Averages
72128	87120	67645	71757	Standard Deviation
0.82	1.02	0.75	0.87	% Difference

Calibrations Used for this Depth/Depth Run Came From the Averages Above.

Time	Device Measured Depth	Tape Measured Depth	Air °F	H ₂ O °F
10:45:40	0.34	0	90	82
10:59:12	5.51	5		
11:09:11	10.43	10	91	
11:18:04	15.28	15		81
11:23:20	19.48	20	92	
11:26:36	24.28	25		80

Air temp increase with H₂O Temp decrease
Helped in this case to maintain a
Reasonable Device Measurement to
Tape Measured Depth Value within norms.

Full Shade

-Refers to the entire setup of the device and the water holding tank in fully shaded conditions.
-The calibrations in full shade conditions rise continually at a steady slow pace assumed to be due to the gradual rise of both the water and air which can only be measured in 1°F increments on the thermometers used. Both upper and lower calibration values

Partial/Mostly Shaded

-Refers to the setup of the device and the water holding tank having some direct sun to any not specified part.
-The calibration in partial shade conditions moves up and down a bit as the upper loadcell may have a few moments in direct sun. Most of the variation is assumed to be due to the upper loadcell conditions as the temperature of the water continued to be stable. It is still to be assumed the water and air temperatures vary by 1°F and are in this case only increasing with the outdoor conditions.

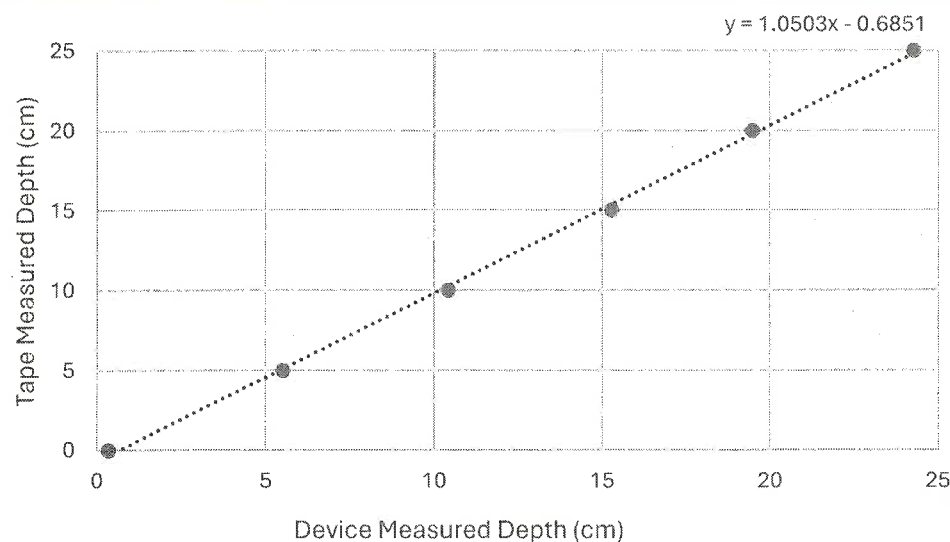
Full Sun

-Refers to the entire device and water holding tank in full sun.
-In full sun the calibrations jumped up rapidly as the temperature of the upper loadcell exceeded the air temperature by several degrees. The effect on the calibrations was instantly recognizable.

Umbrella Shade

-Refers to the upper loadcell being in the shade and the rest of the device and water holding tank in full sun to partial sun to full shade depending on the outdoor conditions.
-It seemed trivial to shade the water as it is not possible to instantly change water temperature even in full sun with the amount of water being used. The water temperature did continue to rise but at about the same rate as previously observed.
- It was observed that the temperature increases to the upper loadcell dropped very quickly after shading was implemented and the calibration values of the lower load cell returned to the previous rate of increase while the upper load cell remained slightly elevated and continued to climb from there with roughly the same rate of increase.

Water Temperature
Decreased with the
Sun No longer Shining
directly into the
Holding tank and
Adding Colder H₂O to increase
water level.
This Day ended Up at
over 100°F so the
Air Temperature
was Rapidly Increasing
even in the morning
and shade.

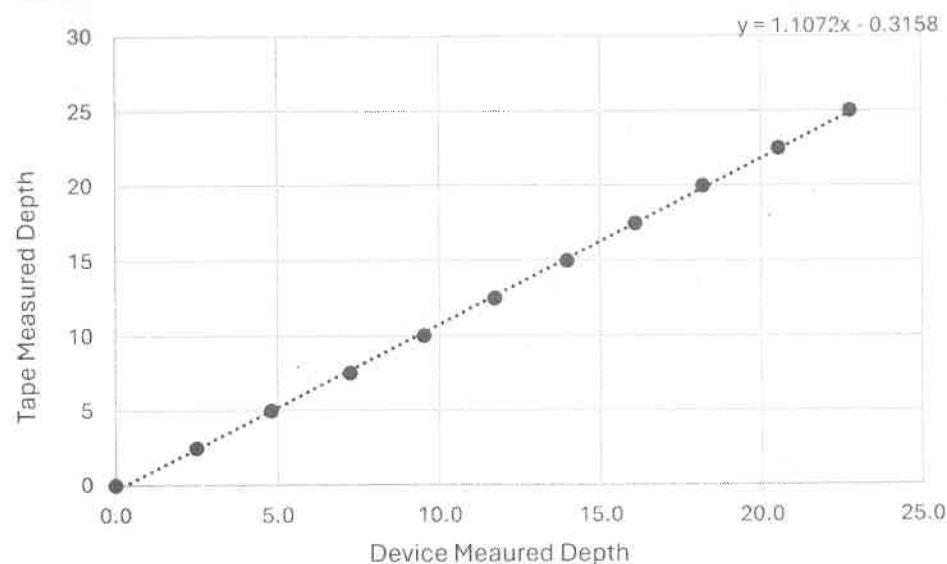


Perfection is a Slope of 1 so this slope ~~helps~~ from Air Temp increase and Water Temp Decreasing Shows the Balance of the Upper + Lower Lead Cells in this Case Came out to Normalized

Time	Device Measured Depth	Tape Measured Depth	Air °F	H ₂ O °F
15:01:26	-0.01	0	97	79
15:03:27	2.5	2.5		
15:05:42	4.8	5		
15:07:42	7.24	7.5		78
15:11:57	9.52	10	98	
15:15:58	11.72	12.5		
15:18:43	13.97	15		
15:21:28	16.08	17.5		
15:24:13	18.19	20		
15:27:44	20.54	22.5		77
15:30:14	22.76	25		

← PM Run Was done Using the Same Calibration as the AM Run

The Water in this Sat in the Shade so the Temp of the Water decreased from the AM Run where the Water Spent a good Part of the Morning in the Sun directly.



Using the AM Calibration The Slope increased from the Morning Run because due to the Temperature differences the Calibration was no longer as Accurate for The PM Run.

Water For This data was left to sit for several hours as water from the hose was added the temperature of the water decreased to match the temperature of the water from the hose. The water sat long enough to have tiny air bubbles over every surface. When I wiped the bubbles the device measured depth slightly increased indicating the air bubbles increased the total liquid capacity. Water was filled to zero level prior to removing the bubbles. The bubbles may indicate evaporation.

→ This led me to do a Full PM Calibration Run that Shows the AM / PM calibration Values are Very Different.

Calibration 6/26 Air 97°F : Water 79°F : Start 3:48 pm

Upper Tare	Lower Tare	Upper750g	Lower750g	Air °F	H ₂ O °F
8823768	8823768	9053398	8327226	97	79
8835687	8677533	9063788	8337894		
8842616	8684630	9072790	8348620	98	
8855389	8698018	9086341	8362494		
8865728	8708631	9099083	8375682		
8874884	8718051	9110943	8387321		
8894006	8737907	9131301	8407987		
8903541	8747720	9138654	8415470		
8911471	8755906	9148741	8425492		
8915071	8759856	9154522	8431779		
8919410	8764192	9158014	8434297		
8923144	8767996	9160101	8435968		
8926126	8771073	9164172	8441991		
8937697	8783167	9173553	8450473		
8938336	8783724	9177863	8455133		
8942769	8788268	9178792	8455384		
8943663	8789130	9176497	8454007		
8941592	8786889	9175303	8451545		
8940066	8785264	9174509	8450804		
8936969	8781967	9172119	8447719		
8938698	8783714	9175546	8451730		
8940326	8785358	9175061	8450823		
8940005	8784937	9173904	8448916		
8935373	8780064	9169984	8444823		
Air 98°F : Water 79°F : Finish 4:28 pm					
8909431	8760323	9144374	8420566	Averages	
37753	36780	39711	40179	Standard Deviation	
0.4	0.4	0.4	0.5	% Difference	

The lower Load Cell Values are On par with what the Am Values were and Climbed in the Same way as Bm. The Upper Load Cell Values are much higher than the morning Temp Values with the exception of when the Upper load cell was in the Full Sun in the morning.

The Overall averages for lower is Close to Am Values for Upper Values are much higher than in the morning.

The Water Temps were Very near (the Same) and the Air Temp Was 10°F higher so this is consistent with expectations that Temperatures Cause Great Variation in Load Cell Consistency.



How Temperature Changes Effect Load Cell Accuracy

Load Cell Zero Drift is the Change in zero load output of the load cell. It is a percentage of the rated load output per $^{\circ}\text{C}$. In Production the load cell is in controlled conditions and Balanced Zero drift is checked to be in required specifications. In Use Zero drift is caused by a combination of factors associated with the components that form the Bridge. Variations in temperature distribution over the load cell body can cause Zero drift greater than data sheet specifications. Error due to Zero drift can be large in application where large changes in temperature occur. The Error, expressed as Force, is independent of the applied load so when using forces at the low end of the load rating the effect is greater. Zeroing out the load cell at the Temperature of Operation can remove Error which in this Experiment is not possible due to ever changing environmental conditions.

Load Cell Span Drift is the change in output for an applied load, expressed as a percentage of the applied load output per $^{\circ}\text{C}$. The load cell body can be considered as a spring with a stiffness defined by Young's Modulus for the spring material. Young's Modulus decreases with rising Temp causing an increase in the load cell output for an applied load, meaning the Resistance for the load cell increases with rising temperature in the bridge circuit. The Span drift of the load cell and the temperature characteristics of the bridge resistors are both non-linear, meaning compensation for the load cell span drift with bridge resistors is not perfect and Error due to Span drift can be large when there are large temperature changes. The error, expressed as a Force is dependent on the applied load so its effects remain the same when forces are only a small % of the rated load need to be measured. If an application only requires load measurements at a relatively static high Temp the span drift ~~error~~ error can be reduced by calibrating at that temperature. This is not possible in this experiment due to the nature of the device having one load cell in rapidly and greatly changing air temps and the other in relatively consistent water temps which both push against each other.

Summary of the Variation in Calibration Values

Calibration Averages in Different Air and Water Temperature Conditions									
Upper Tare	Lower Tare	Upper 750g	Lower 750g	Upper Δ	Lower Δ	Difference In Δ 's	Tare Δ	750g Δ	Difference In Δ 's
Room Temperature Air and Water									
8598996	8425027	8805932	8060228	206936	364799	157863	173969	745704	571735
Air Temperature Below 90°F Water Temperature not accounted for									
8703847	8530871	8867082	8116501	163235	414370	251135	172976	750581	577605
Air Temperature Above 90°F Water Temperature not accounted for									
8898924	8736937	9111956	8381061	213032	355876	142844	161987	730895	568908
Air 70°F : Water 67°F									
8580833	8406720	8778677	8027390	197844	379330	181486	174113	751287	577174
Air 70°F : Water 70°F									
8611950	8434374	8895047	8140284	283097	294090	10993	177576	754763	577187
Air 70°F : Water 75°F									
8593301	8420152	8774494	8025225	181193	394927	213734	173149	749269	576120
Air 70°F : Water 85°F									
8544402	8363340	8733183	7976496	188781	386844	198063	181062	756687	575625
Start Air 84°F Water 77°F : Finish Air 88°F Water 80°F									
8674412	8497124	8903864	8161089	229452	336035	106583	177288	742775	565487
Start Air 85°F Water 78°F : Finish Air 98°F Water 79°F									
8783275	8589892	9014769	8279877	231494	310015	78521	193383	734892	541509

Because Both Load cells are Inflexibly Bound Together the measurement of one is the exact opposite of the other. This means if one is off the other is off exactly that amount. These Load cells Read Opposite of Each other so if one goes high the other will Read Low in order to maintain equilibrium. If overall depth measurement is 20cm Together the Load cells will have to equal that value Regardless of how the average is achieved.

When Temperature Changes in Either the Upper or Lower Load cell the measurement of the effected Load cell Varies from what actual would be and the other Load cell changes to make up the difference

By Holding the Upper Load cell at Constant Air Temperature and Varying the Lower Load cell Temperature the Calibration Values Vary as do the depth measurements for Overall, Upper Reading and Lower Reading. If not Carefully aware of this One could easily Overlook the resulting measurements inaccuracy and assume the Upper + Lower Readings to be Correct because the Overall depth Measurement calculates to correct.

Determining How Calibrations Change when Air Temp Is Held Constant and H₂O Temp Changes

Calibration 6/27 Air 70°F : Water 67°F			
Upper Tare	Lower Tare	Upper 750g	Lower 750g
8538079	8362150	8782642	8030986
8585095	8411167	8780854	8029843
8585403	8411429	8780189	8028914
8585246	8411309	8778179	8027062
8584087	8410077	8778465	8025815
8584830	8410873	8778009	8026680
8583659	8409643	8777095	8026125
8585679	8411779	8776581	8026032
8585170	8411284	8777240	8026171
8584393	8410461	8777747	8026294
8587520	8413747	8778445	8027365
8580833	8406720	8778677	8027390

Calibration 6/27 Air 70°F : Water 70°F			
Upper Tare	Lower Tare	Upper 750g	Lower 750g
8621041	8443865	8897930	8144208
8616909	8439605	8896178	8141792
8613125	8435602	8895305	8140747
8614178	8436727	8894857	8139372
8610297	8432653	8893726	8139862
8608900	8431221	8894257	8139232
8608350	8430580	8894395	8140167
8608419	8430678	8894656	8138481
8606329	8428434	8894119	8138695
8611950	8434374	8895047	8140284

Calibration 6/27 Air 70°F : Water 75°F			
Upper Tare	Lower Tare	Upper 750g	Lower 750g
8589661	8416604	8773669	8023650
8597089	8424341	8774269	8024759
8598516	8425773	8775428	8025772
8590827	8417620	8774984	8025180
8593644	8420459	8775119	8025739
8594316	8421106	8775181	8026012
8593760	8420517	8774484	8025725
8587414	8413877	8774195	8024449
8590332	8416874	8773537	8025223
8597447	8424349	8774074	8025738
8593301	8420152	8774494	8025225

Calibration 6/27 Air 70°F : Water 85°F			
Upper Tare	Lower Tare	Upper 750g	Lower 750g
8538574	8357194	8738832	7982204
8548697	8367636	8737198	7980836
8548693	8367597	8733986	7977605
8544340	8363046	8730789	7973787
8541355	8360172	8731211	7974618
8546311	8365450	8733494	7976078
8545758	8364817	8731033	7974329
8541284	8360174	8732312	7976112
8545445	8364626	8731535	7974572
8543566	8362692	8731436	7974819
8544402	8363340	8733183	7976496

Calibration Values Used Air 70° F : Water 70° F			
Upper Tare	Lower Tare	Upper 750g	Lower 750g
8611950	8434374	8895047	8140284
Tape Depth	Total Depth	Lower Depth	Time
0	-0.02	0.67	15:24:55
5	4.74	6.19	15:26:55
10	9.51	10.70	15:29:55
15	14.68	15.47	15:32:26
20	19.59	19.48	15:34:41

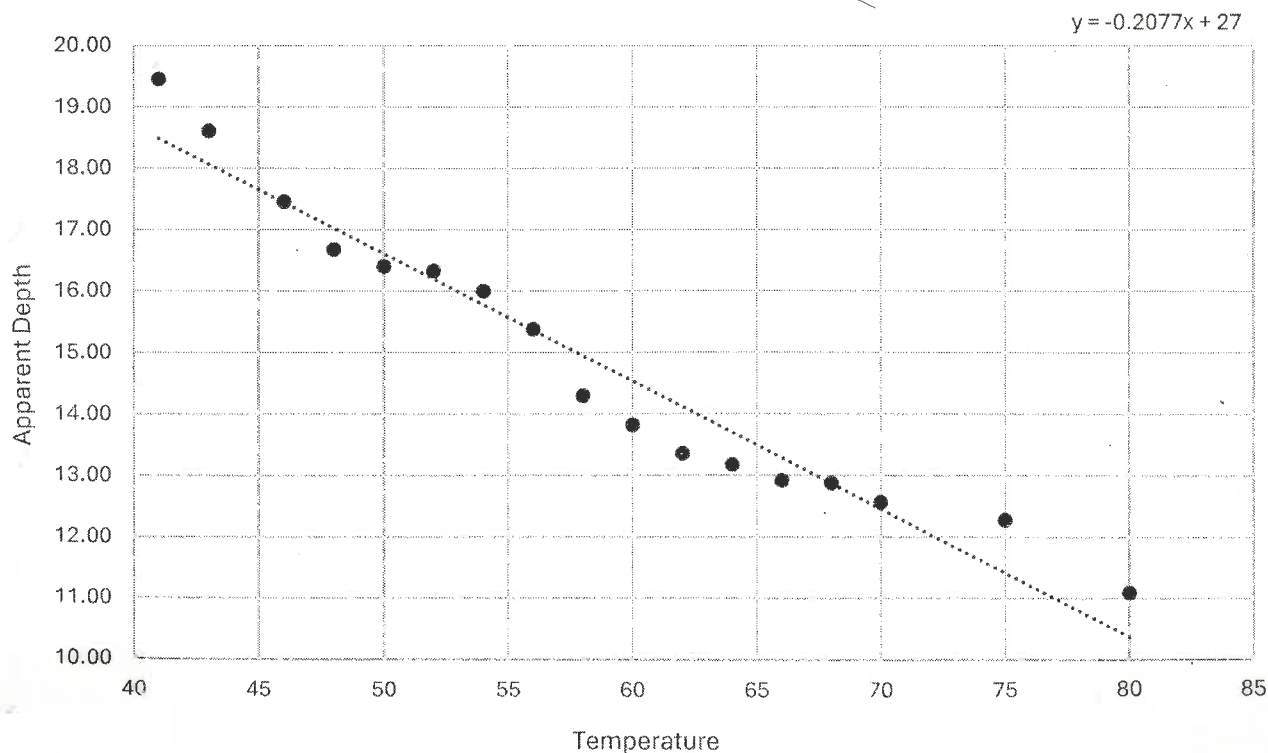
Calibration Values Used Air 70° F : Water 70° F			
Upper Tare	Lower Tare	Upper 750g	Lower 750g
8611950	8434374	8895047	8140284

20cm Depth by Tape Measure Air 70° F			
H ₂ O Temp ° F	Total Depth	Lower Depth	Upper Depth
41	18.52	19.45	17.59
43	18.56	18.61	18.51
46	18.60	17.46	19.74
48	18.63	16.68	20.58
50	18.66	16.40	20.92
52	18.69	16.33	21.05
54	18.73	16.00	21.46
56	18.76	15.38	22.14
58	18.80	14.30	23.30
60	18.83	13.82	23.84
62	18.86	13.36	24.36
64	18.93	13.18	24.69
66	18.94	12.92	24.96
68	19.02	12.88	25.16
70	19.07	12.57	25.57
75	19.18	12.28	26.08
80	19.26	11.10	27.43



Holding Air (Upper Load cell) Constant Temperature.

- * When H₂O Temp Changes (Lower Load cell) to higher temperatures
 - Due to Young's Modulus the lower Load cell Becomes more Flexible indicating a false ~~higher~~ Force Reading
 - Even with Upper Load cell at Constant Temp The Upper Reading Responds with an Opposite of Lower Load cell Value that indicates a False higher Force Value as well.



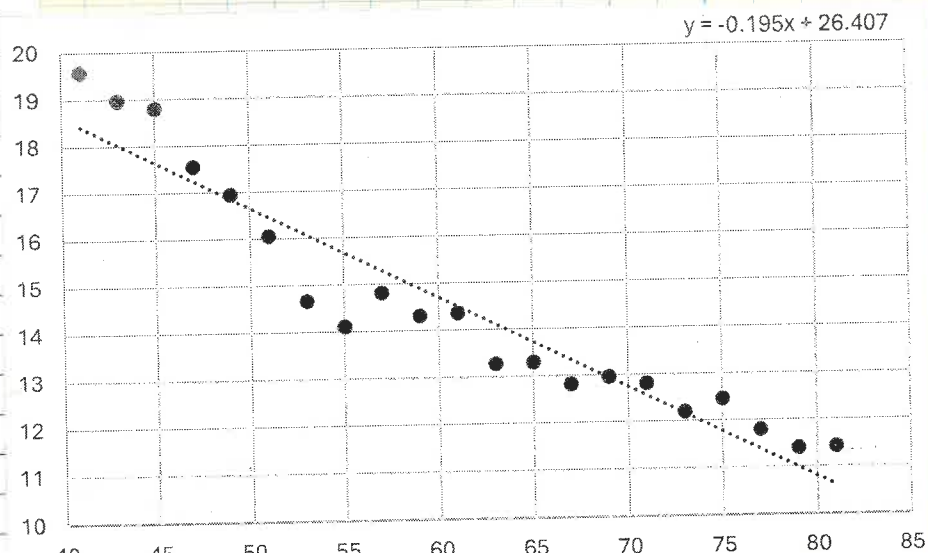
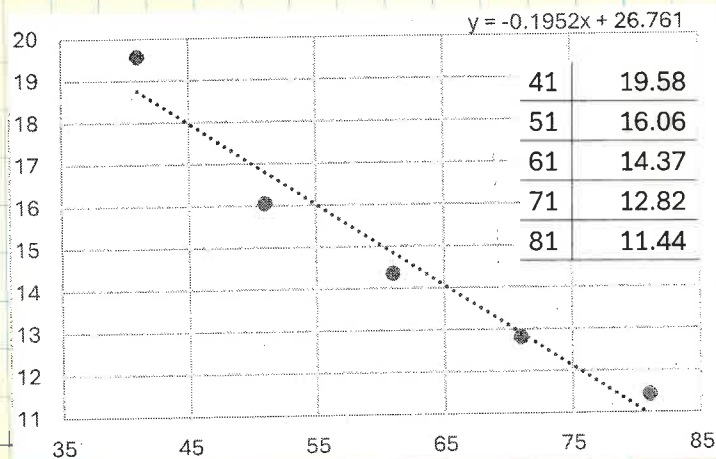
When one goes UP Regardless of cause the other must go down and therefore neither upper nor lower readings can/will be accurate even if using Calibrations Set for the Temperature that was held constant

Calibration Values Used Air 70° F : Water 70° F

Upper Tare	Lower Tare	Upper 750g	Lower 750g
8611950	8434374	8895047	8140284

20cm Depth by Tape Measure Air 70° F

H ₂ O Temp ° F	Total Depth	Lower Depth	Upper Depth
39	17.90	21.48	14.31
41	17.93	19.58	16.29
43	18.13	18.97	17.29
45	18.19	18.80	17.57
47	18.24	17.55	18.93
49	18.33	16.95	19.72
51	18.36	16.06	20.66
53	18.39	14.67	22.10
55	18.48	14.12	22.85
57	18.56	14.82	22.31
59	18.59	14.32	22.85
61	18.60	14.37	22.83
63	18.65	13.28	24.03
65	18.70	13.30	24.09
67	18.74	12.83	24.64
69	18.76	12.97	24.54
71	18.92	12.82	25.03
73	18.98	12.21	25.75
75	19.01	12.47	25.55
77	19.03	11.81	26.25
79	19.05	11.41	26.70
81	19.11	11.44	26.78

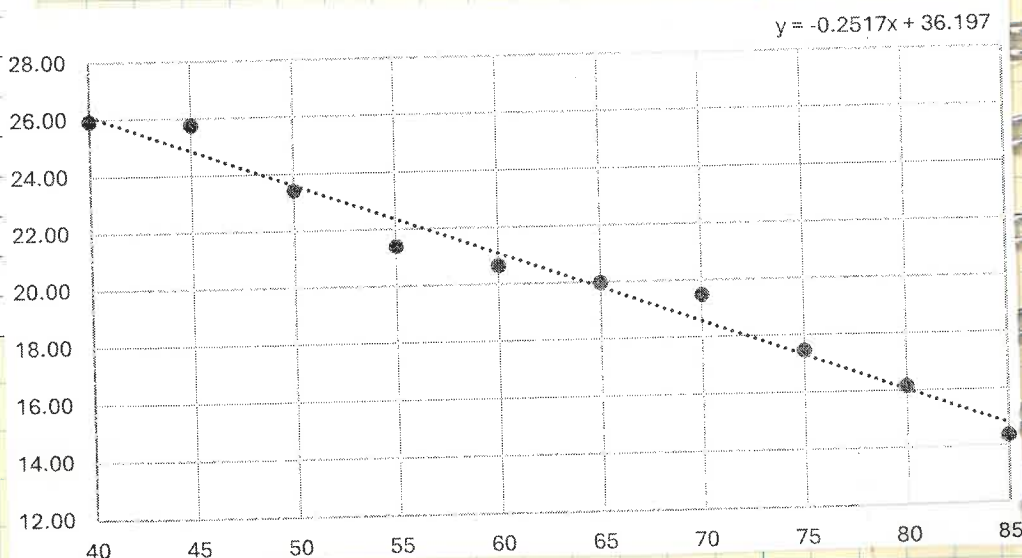


Calibration Values Used 70° F : Water 70° F

Upper Tare	Lower Tare	Upper 750g	Lower 750g
8611950	8434374	8895047	8140284

20cm Depth by Tape Measure Air 70° F

H ₂ O Temp ° F	Total Depth	Lower Depth	Upper Depth
40	18.32	27.21	9.43
45	18.43	25.93	10.94
50	18.62	23.36	13.88
55	18.74	21.57	15.91
60	18.76	20.27	17.25
65	18.86	20.01	17.72
70	19.06	19.43	18.77
75	19.26	17.56	20.96
80	19.37	15.87	22.87
85	19.44	13.63	25.25



Conclude that a Single Loadcell Placed at Bottom of Pipe is (in H₂O) Best Choice For Limiting Temperature Swing effecting Measurements.